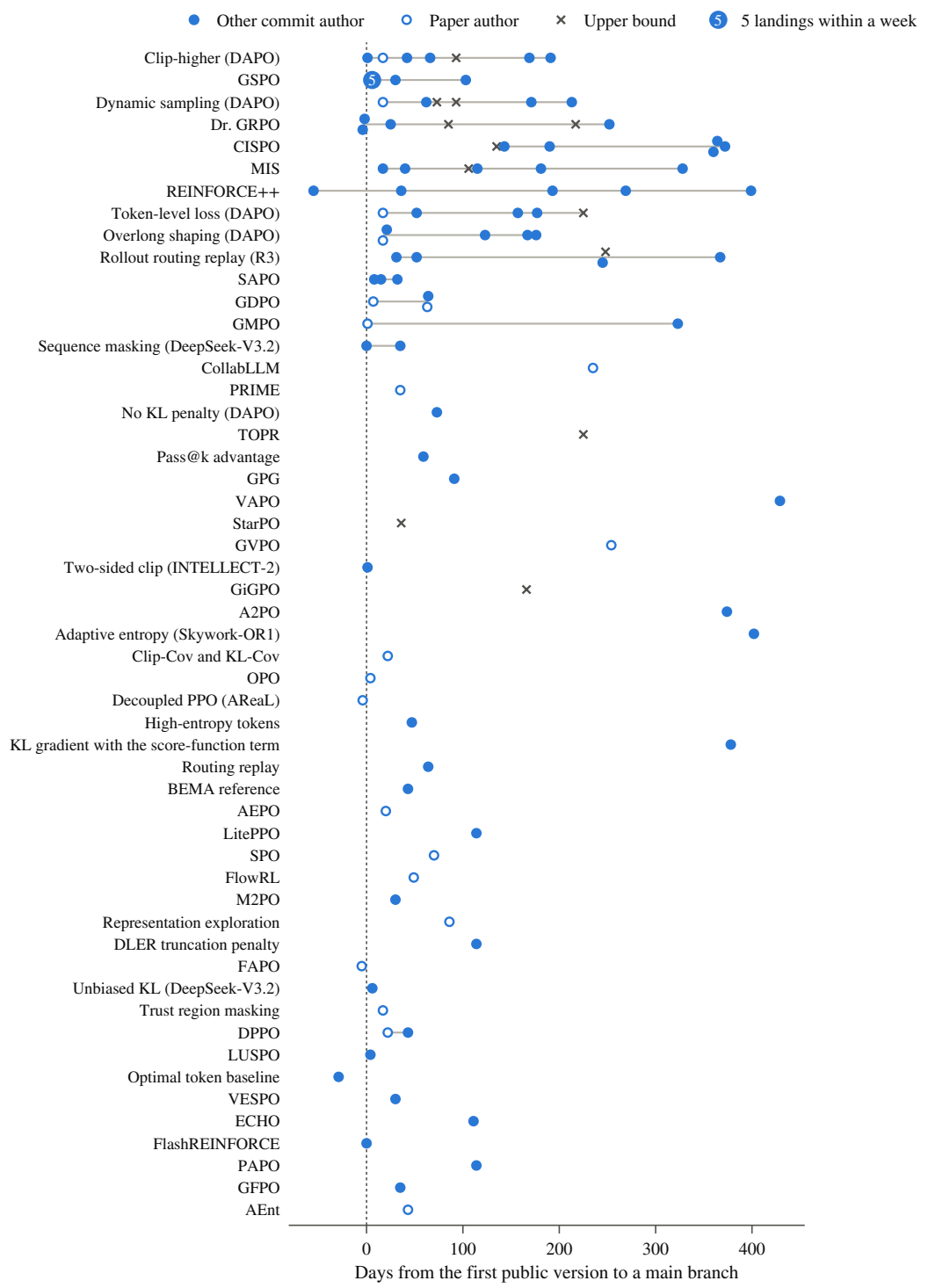
Other commit author
Paper author
Upper bound
5 landings within a week
Clip-higher (DAPO)
GSPO
Dynamic sampling (DAPO)
Dr. GRPO
CISPO
MIS
REINFORCE++
Token-level loss (DAPO)
Overlong shaping (DAPO)
Rollout routing replay (R3)
SAPO
GDPO
GMPO
Sequence masking (DeepSeek-V3.2)
CollabLLM
PRIME
No KL penalty (DAPO)
TOPR
Pass@k advantage
GPG
VAPO
StarPO
GVPO
Two-sided clip (INTELLECT-2)
GiGPO
A2PO
Adaptive entropy (Skywork-OR1)
Clip-Cov and KL-Cov
OPO
Decoupled PPO (AReaL)
High-entropy tokens
KL gradient with the score-function term
Routing replay
BEMA reference
AEPO
LitePPO
SPO
FlowRL
M2PO
Representation exploration
DLER truncation penalty
FAPO
Unbiased KL (DeepSeek-V3.2)
Trust region masking
DPPO
LUSPO
Optimal token baseline
VESPO
ECHO
FlashREINFORCE
PAPO
GFPO
AEnt
0
100
200
300
400
Days from the first public version to a main branch